

Εὐκλείδης ὁ Ἀλεξανδρεύς

Euclid of Alexandria and the Elements of Mathematics

eul_aid: iaw

Period: Ἑλληνιστική [Hellenistic](#)
Dialect: Κοινή τεχνική [Technical Koine](#)
Domain: Μαθηματικά [Mathematics](#)
Format: Πραγματεία [Treatise](#)

Euclid was an ancient Greek mathematician active in the early Hellenistic period, around 300 BCE. He worked in Alexandria, Egypt, during the reign of King Ptolemy I, where he founded a mathematical school. Very few reliable details about his personal life, birthplace, or education are known.

He is the author of several major mathematical treatises. His most famous and influential work is the *Elements*, a systematic compilation of the geometric knowledge of his time. Other significant surviving works include the *Data*, on geometric problems; the *Phaenomena*, on spherical astronomy; and the *Optics*, on perspective. The authorship of some works, like the *Catoptrics* on mirrors, is sometimes disputed. He also wrote several important works that are now lost, such as the *Conics* and *Porisms*.

Euclid's historical importance stems from his synthesis of existing mathematics into a rigorous logical system. The *Elements* is one of the most successful textbooks ever written, serving as the foundational text for geometry for over two thousand years. Its method of starting from axioms and definitions and proceeding by deductive proof became a model for scientific and philosophical exposition. Translated into Arabic and later Latin, it was central to education from the medieval period through the Renaissance and profoundly influenced later thinkers like Isaac Newton. While much of the content was not original, his achievement was in its masterful organization and proof.

Works (17)

- Κατοπτρικά · *Catoptrics*, Theon Edition? · 32 passages
- Δεδομένα · *Data* · 95 passages
- Δεδομένα καὶ Ἄλλαι Ἀποδείξεις · *Data Other Demonstrations* · 23 passages
- Στοιχεῖα Ἄλλαι ἀποδείξεις (βιβλ. ι') · *Elements, Alternative Demonstrations (Book X)* · 28 passages
- Στοιχεῖα Ἄλλαι ἀποδείξεις (βιβλ. ια') · *Elements, Alternative Demonstrations (Book XI)* · 13 passages
- Στοιχεῖα Ἄλλαι ἀποδείξεις (βιβλ. α'–γ') · *Elements, Alternative Demonstrations (Books I–III)* · 8 passages
- Στοιχεῖα Ἄλλαι ἀποδείξεις (βιβλ. ε'–ζ') · *Elements, Alternative Demonstrations (Books V–VI)* · 10 passages
- Στοιχεῖα (βιβλ. α'–ιγ') · *Elements, Books I–XIII* · 624 passages
- Στοιχεῖα (βιβλ. ια'–ιβ', ἀποσπάσματα) · *Elements, Books XI–XII (Supplement)* · 30 passages
- Ἐπίγραμμα · *Epigram* · 3 passages
- Ἀποσπάσματα · *On Divisions of Figures* · 50 passages
- Ὀπτικά · *Optics* · 63 passages
- Φαινόμενα · *Phenomena* · 26 passages

- Φαινόμενα (Ἀποδείξεις Ἀναθεωρήσεως Β) · Phenomena, Demonstrations of Another Recension B · 6 passages
- Φαινόμενα (Ἀναθεώρησις Β) · Phenomena, Recension B · 15 passages
- Ἐπισκόπησις Ὀπτικῶν ὑπὸ Θεωνος · Review of Optics by Theon · 52 passages
- Κανόνος Τομή · Section of the Canon · 21 passages